

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460120.9019 +/- 0.0075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.214 +/- 0.063
Transit depth (Rp/Rs)^2	4.56 +/- 2.69 [%]
Orbital Inclination (inc)	88.06 +/- 2.76
Airmass coefficient 1 (a1)	0.71 +/- 0.69
Airmass coefficient 2 (a2)	0.33 +/- 0.22
Scatter in the residuals of the lightcurve fit is	143.49 %
Best Comparison Star	None
Optimal Aperture	1.58
Optimal Annulus	3.76
Transit Duration (day)	0.085 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.214 ± 0.063 vs 0.1646 (deviation 0.78σ)
Duration fit vs published	0.085 d vs 0.0758 d (deviation 0.83σ)
Mid-time O–C	16.6 min
Depth SNR	3.4
Residual scatter	143.49 %

Light curve

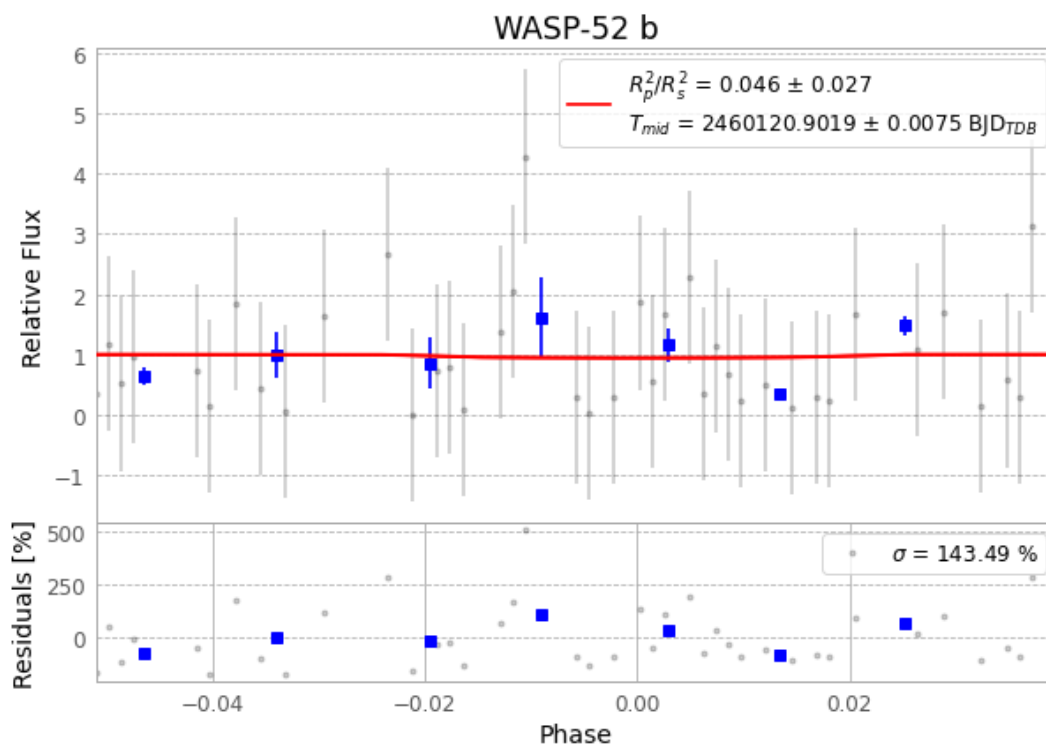


Figure 1 — FinalLightCurve_WASP-52 b_2023-06-25.png (SHA-256 b567a99f913dcc12...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [171, 372], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 dba240373a7c795e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 72f1d3a

Data provenance

77 FITS frames (51 MB), WASP-52230625072715.FITS → WASP-52230625111515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-230625.log (666a7052eb3d...), FinalLightCurve_WASP-52 b_2023-06-25.png (b567a99f913d...), FinalLightCurve_WASP-52 b_2023-06-25.csv (f4c686f4762e...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 07:38Z.